

A Comparative Analysis of Optimization Trajectories and Convergence Properties.

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Abstract—The choice of the optimization algorithm determines training efficiency and generalization of deep neural networks. While methods such as Stochastic Gradient Descent and Adam have become field standards, recent works such as Lion [Che+25] and Muon [Jor+24] promise better convergence properties in practice. In this paper we present a comparative study of the optimization trajectories produced by Stochastic Gradient Descent, Adam, Lion and Muon applied to Multi-Layer Perceptron on the Fashion-MNIST dataset [XRV17]. To this end, we use a linear interpolation method: for parameters θ_t before and θ_{t+1} after an epoch, we evaluate the loss function at $N = 50$ points along the connecting vector, allowing us to empirically verify convexity and smoothness properties along the trajectory.

Furthermore, we evaluate the convergence points θ by testing the condition:

$$f(\theta) \leq f(x) + \nabla f(\theta)^\top (\theta - x),$$

for points x within a neighborhood $B_\epsilon(\theta)$.

We use constant learning rate, and batch-size found by grid-search and compare them with predicted theoretical values, all other hyperparameters are set to default.

I. INTRODUCTION

The choice of the optimizer is central to the process of training deep neural networks, influencing not only the speed of convergence but also the ability of the model to generalize. There is a vast number of available optimizers that can be applied to train a deep neural network, [SDH21] identified over one hundred unique algorithms, yet only two have dominated the scene: Stochastic Gradient Descent (SGD) [RM51] and Adam [KB15]. The Lion optimizer, which was developed based on symbolic search for update equations, is able to achieve similar levels of effectiveness without having problems with large amounts of memory usage. Furthermore, the Muon optimizer, developed initially to speed up the NanoGPT training, has shown better convergence rates.

Empirical comparisons typically report aggregate metrics such as final test accuracy or training loss curves. Less studied is the geometry of the optimization trajectory itself: whether successive parameter updates traverse locally convex regions of the loss landscape, how smooth the loss appears along each step, and whether convergence points satisfy local optimality conditions. Our analysis shows behavioral differences between optimizers that aggregate metrics alone may not reveal.

II. MODELS AND METHODS

We fit an MLP on the Fashion MNIST dataset which includes 10 different classes for image classification. This dataset contains a total of 60,000 gray-scale images of size 28x28. Our model is composed of two hidden layers of width 256 and 128 respectively, each with ReLU activation, followed by a linear classification layer, giving a total of approximately 234K trainable parameters. The cross-entropy loss is used with no regularization nor weight-decay for simplicity.

The following four optimization algorithms are compared. Stochastic gradient descent updates its parameters using

$$\theta_{t+1} = \theta_t - \eta g_t.$$

Here, g_t represents the stochastic gradient. The Adam optimizer computes the first- and second-moment averages of the gradient and performs bias-corrected adaptive steps per coordinate. The Lion optimizer uses only the sign of a momentum-gradient sum as the update direction:

$$\theta_{t+1} = \theta_t - \eta \text{sign}(\beta_1 m_t + (1 - \beta_1) g_t).$$

The momentum is updated by

$$m_{t+1} = \beta_2 m_t + (1 - \beta_2) g_t.$$

The Muon optimizer implements the Nesterov-type momentum in the direction of steepest descent in terms of the spectral norm and approximates it by performing the Newton-Schulz orthogonalization of the gradient matrix.

The hyperparameters used for our analysis are found by performing a grid search over learning rates in $\{10^{-4}, 10^{-3}, 10^{-2}\}$ and batch sizes in $\{32, 64, 128\}$ and selecting the combination that activation the lowest training loss for each optimizers after epoch 20. All other hyperparameters are kept at their published defaults.

For each optimizer, the parameter vectors θ_t and θ_{t+1} are stored at each epoch for a total of 5 epochs. The loss over the training set is computed at the points along the line

$$\theta(\alpha) = (1 - \alpha)\theta_t + \alpha\theta_{t+1}$$

for $\alpha \in \{0, 1/N, \dots, 1\}$ and $N = 50$.

This allows us to analyze local approximation of the loss along the “average” trajectory.

Furthermore, we verify the first-order characterization of convexity by sampling $K = 50$ points in the ball $B_\varepsilon(\theta)$ with $\varepsilon = 10^{-2}$, and checking,

$$f(\theta) \leq f(x) + \nabla f(\theta)^\top (\theta - x),$$

providing an empirical measure of local optimality.

III. RESULTS

The selected learning rates and batch-sizes along with final train loss and test accuracy after 20 epochs are shown in Table I.

TABLE I
SELECTED LEARNING RATES AND FINAL PERFORMANCE AFTER 20 EPOCHS.

Optimizer	LR	Batch Size	Train Loss	Test Acc. (%)
SGD	10^{-2}	32	0.2383	88.38
Adam	10^{-3}	64	0.1426	88.66
Lion	10^{-4}	128	0.1269	88.52
Muon	10^{-3}	64	0.0600	90.01

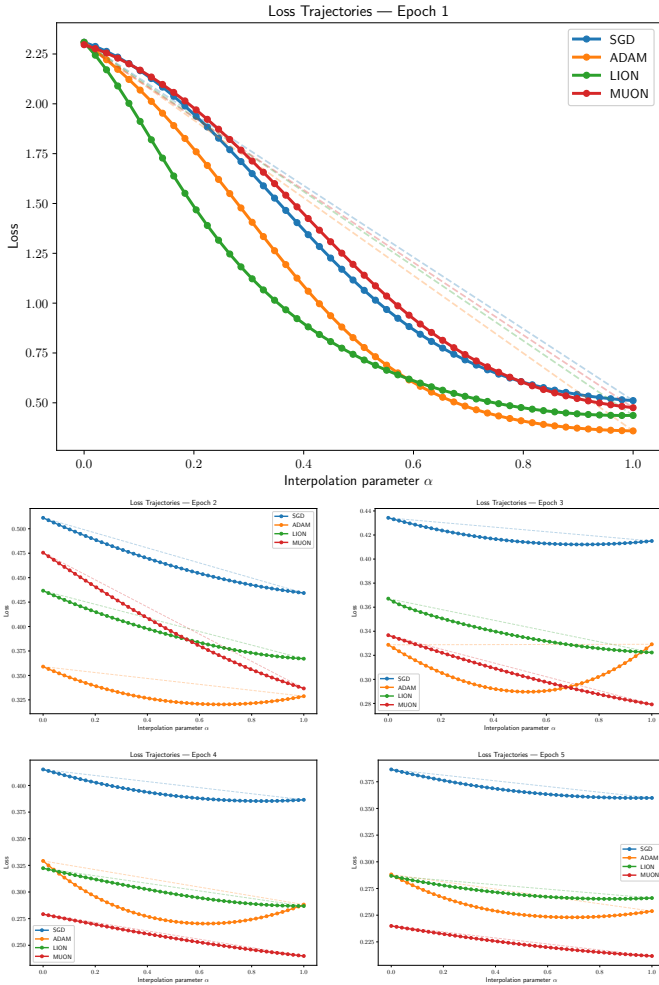


Fig. 1. Loss evaluated along the linear interpolation $\theta(\alpha) = (1-\alpha)\theta_t + \alpha\theta_{t+1}$ for each epoch. Dashed lines show the linear interpolation of endpoint losses.

TODO: Convergence test results.

IV. DISCUSSION

TODO: Update for the new results.

All four optimizers show competitive test accuracy on Fashion-MNIST dataset. The highest test accuracy is achieved by both Adam 87.74% and Lion 87.81%, followed by Muon with 87.29% and SGD with 85.43%. Lion optimizer achieves an accuracy comparable to Adam with a learning rate that is 10x lower than the latter due to the sign-based update producing larger step sizes.

TODO: Update for the new results. Loss Landscape along Update Directions. Fig. 1 illustrates the evolution of loss values along $N = 20$ interpolation points on the line $\theta(\alpha) = (1-\alpha)\theta_t + \alpha\theta_{t+1}$ across all 5 training epochs. At the early stage of the training process (epoch 1), SGD and Muon produce non-convex paths, while Adam and Lion are already showing convex paths. Starting from epoch 2, SGD, Adam, and Lion demonstrate convex, monotonically decreasing paths, while Muon remains non-convex until epoch 4, reflecting the more exploratory nature of its orthogonalized updates.

TODO: Discuss the convergence test results.

V. SUMMARY

TODO: Repeat abstract with results in mind.

REFERENCES

- [Che+25] Lizhang Chen et al. *Lion Secretly Solves Constrained Optimization: As Lyapunov Predicts*. 2025. arXiv: 2310.05898 [cs.LG]. URL: <https://arxiv.org/abs/2310.05898>.
- [Jor+24] Keller Jordan et al. *Muon: An optimizer for hidden layers in neural networks*. 2024. URL: <https://kellerjordan.github.io/posts/muon/>.
- [KB15] Diederik P. Kingma and Jimmy Ba. *Adam: A Method for Stochastic Optimization*. 2015. arXiv: 1412.6980 [cs.LG]. URL: <https://arxiv.org/abs/1412.6980>.
- [RM51] Herbert Robbins and Sutton Monro. “A Stochastic Approximation Method”. In: *The Annals of Mathematical Statistics* 22.3 (1951), pp. 400–407. URL: <https://doi.org/10.1214/aoms/1177729586>.
- [SDH21] Frank Schneider, Felix Dangel, and Philipp Hennig. “Descending through a Crowded Valley — Benchmarking Deep Learning Optimizers”. In: *Proceedings of the 38th International Conference on Machine Learning (ICML)*. 2021. arXiv: 2007.01547 [cs.LG]. URL: <https://arxiv.org/abs/2007.01547>.
- [XRV17] Han Xiao, Kashif Rasul, and Roland Vollgraf. *Fashion-MNIST: a Novel Image Dataset for Benchmarking Machine Learning Algorithms*. Aug. 28, 2017. arXiv: cs.LG / 1708.07747 [cs.LG].